

ANALISIS DE DATOS DE ARTICULO DE CRECIMIENTO CORALINO EN VARADERO

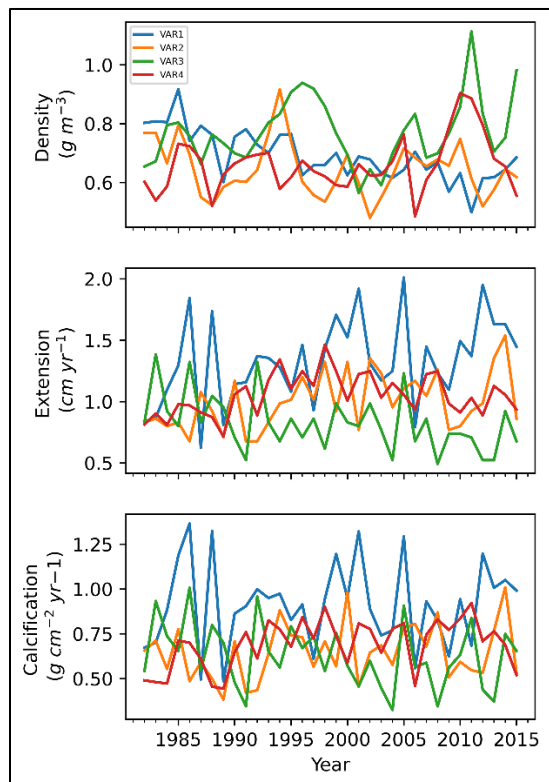


Fig. 1. Tasas de crecimiento de cada núcleo de coral periodo 1982-2015.

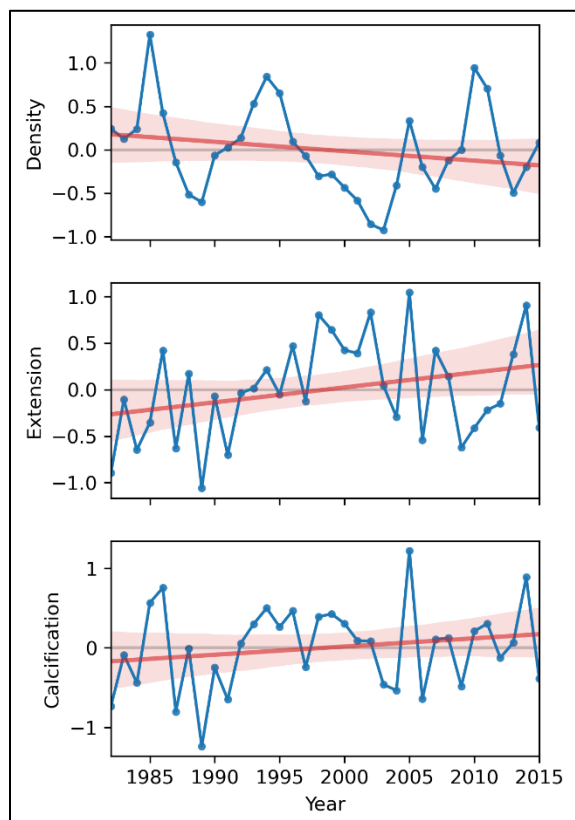
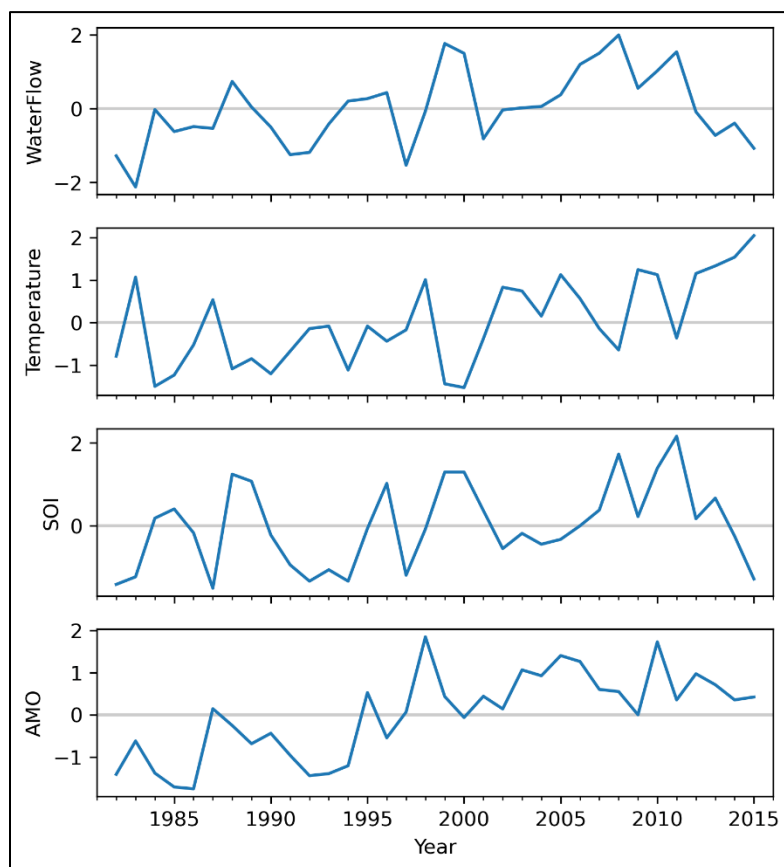
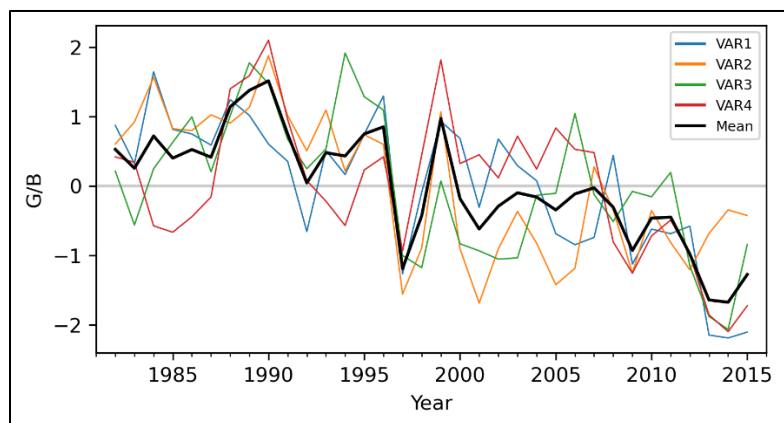


Fig. 2. Tendencia de las tasas de crecimiento normalizado de cada núcleo de coral periodo 1982-2015. Zona sombreada indica intervalo de confianza al 95%



Multiple linear regression results of annual coral growth parameters and luminescence against the annual environmental variables, **period 1982-2015**.

Modelo incluyendo: Water Flow + Temperature + SOI + AMO.

The environmental variables with $p < 0.05$ have significant effect on the response variable.

AIC-based stepwise forward select: La columna AIC indica cual es la mejor variable para incluir en el modelo de acuerdo a cuanto explica en la variable de respuesta.

Variable	Slope	Slope Error	t-stat	p-value	AIC
<i>Density</i>					
Water Flow	0.013	0.015	0.857	0.421	Best variable: AMO
Temperature	0.014	0.014	1.055	0.323	
SOI	0.002	0.014	0.121	0.913	
AMO	-0.029	0.014	-2.023	0.063	
<i>Extension</i>					
Water Flow	0.015	0.041	0.359	0.719	Best variable: AMO
Temperature	0.019	0.038	0.496	0.617	
SOI	0.017	0.037	0.455	0.659	
AMO	0.017	0.040	0.422	0.687	
<i>Calcification</i>					
Water Flow	0.019	0.030	0.628	0.533	No best variable
Temperature	0.024	0.028	0.854	0.406	
SOI	0.019	0.027	0.687	0.490	
AMO	-0.026	0.029	-0.886	0.393	
<i>Luminescence</i>					
Water Flow	0.003	0.004	0.708	0.480	Best variable: Temperature
Temperature	-0.012	0.003	-3.476	0.001	
SOI	-0.004	0.003	-1.115	0.272	
AMO	-0.002	0.004	-0.467	0.650	

Multiple linear regression results of annual coral growth parameters and luminescence against the annual environmental variables, **period 1954-1980**.

Modelo incluyendo: Temperature + SOI + AMO. The environmental variables with $p < 0.05$ have significant effect on the response variable.

AIC-based stepwise forward select: La columna AIC indica cual es la mejor variable para incluir en el modelo de acuerdo a cuanto explica en la variable de respuesta.

Variable	Slope	Slope Error	t-stat	p-value	AIC
<i>Density</i>					
Temperature	0.017	0.011	1.470	0.166	No best variable
SOI	0.017	0.012	1.509	0.155	
AMO	0.001	0.011	0.073	0.942	
<i>Extension</i>					
Temperature	-0.009	0.015	-0.610	0.525	Best variable: AMO
SOI	0.010	0.015	0.672	0.487	
AMO	0.048	0.014	3.448	0.002	
<i>Calcification</i>					
Temperature	0.010	0.013	0.792	0.436	Best variable: AMO
SOI	0.027	0.013	2.076	0.044	
AMO	0.034	0.012	2.840	0.007	
<i>Luminescence</i>					
Temperature	-0.001	0.002	-0.519	0.598	Best variable: AMO
SOI	0.002	0.002	1.180	0.256	
AMO	0.003	0.002	1.871	0.083	

FIGURAS DE LA SERIE DE TIEMPO COMPLETA (USANDO DATOS NORMALIZADOS) – NO HAY ANALISIS ESTADISTICOS, SOLO FIGURAS PARA INTERPRETAR LOS DATOS.

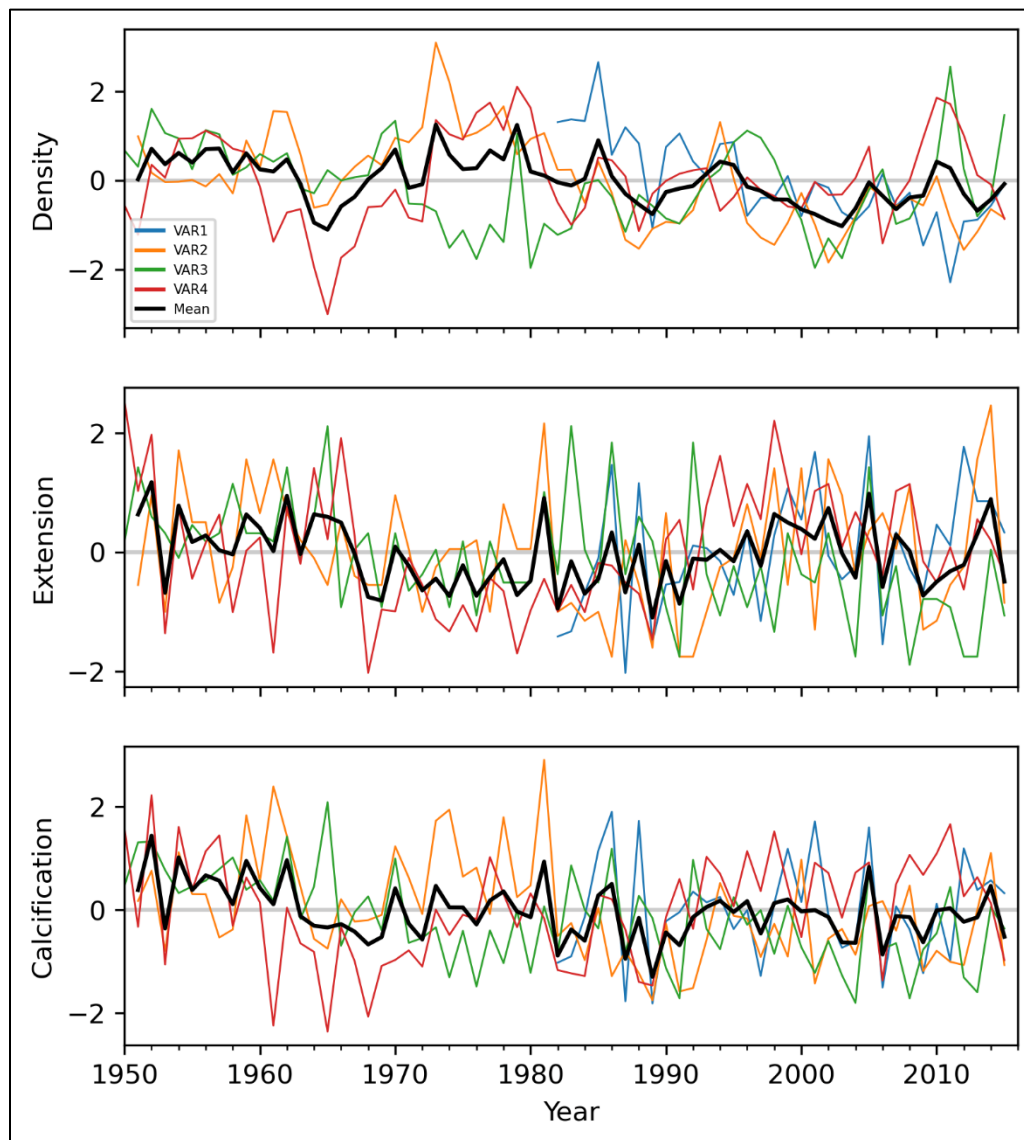
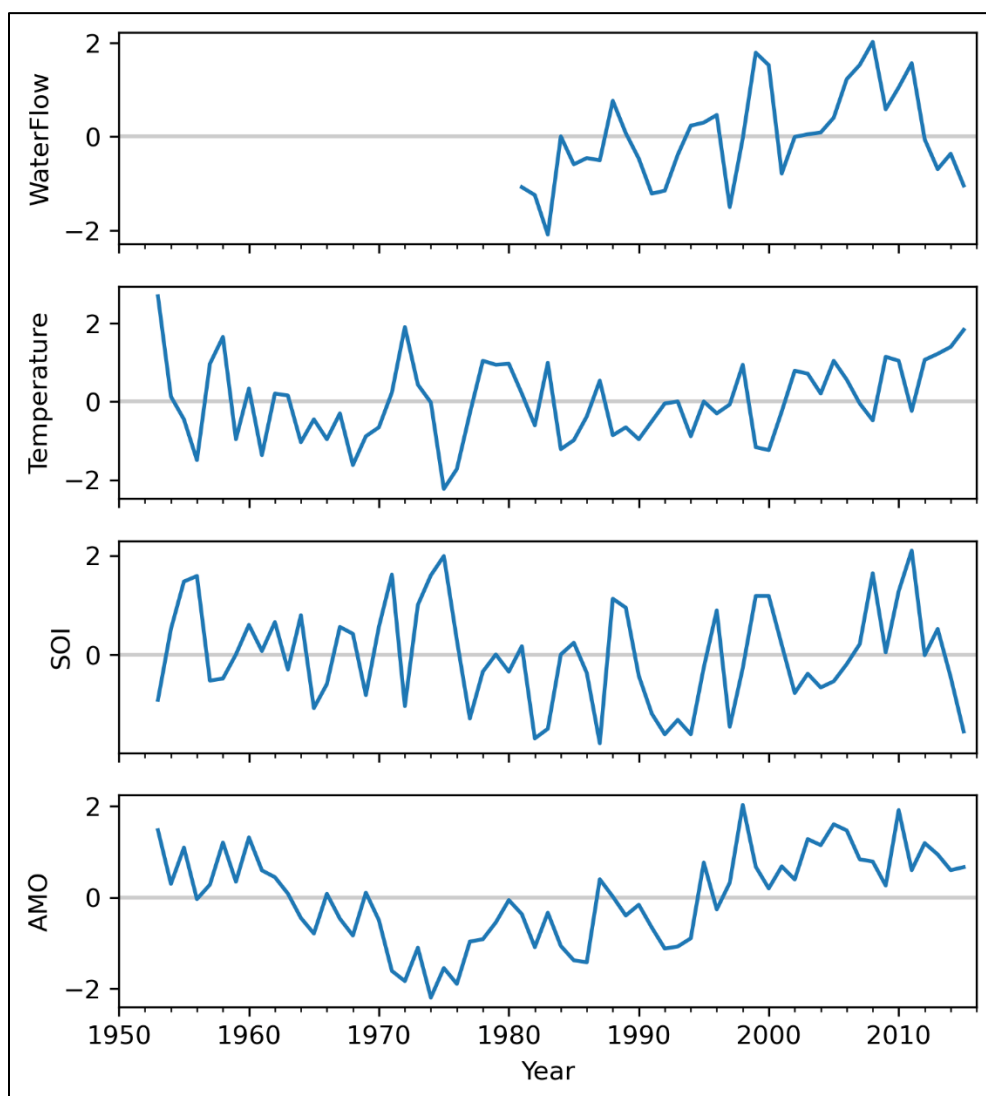
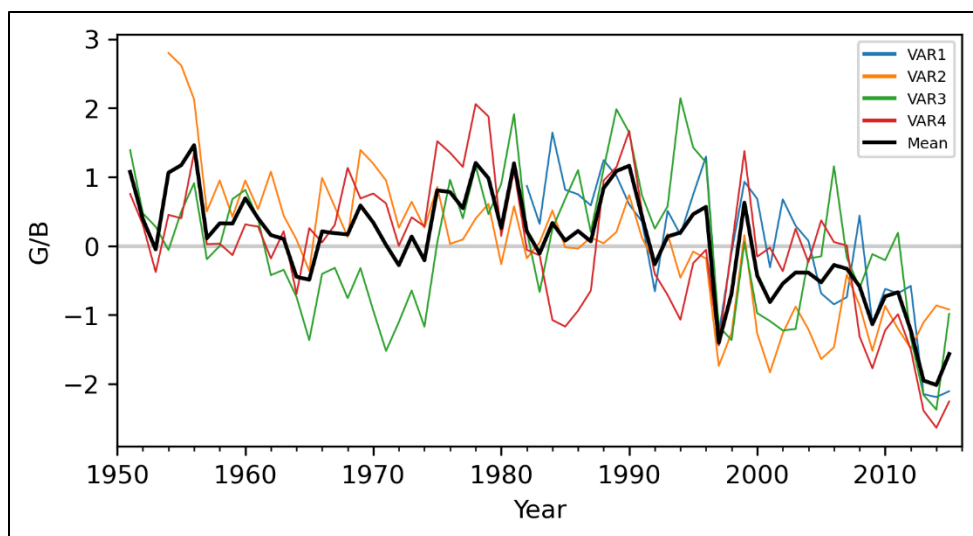


FIG.5. CRECIMIENTO CORALINO NORMALIZADO,
POR CORE Y PROMEDIO GENERAL (n=4)



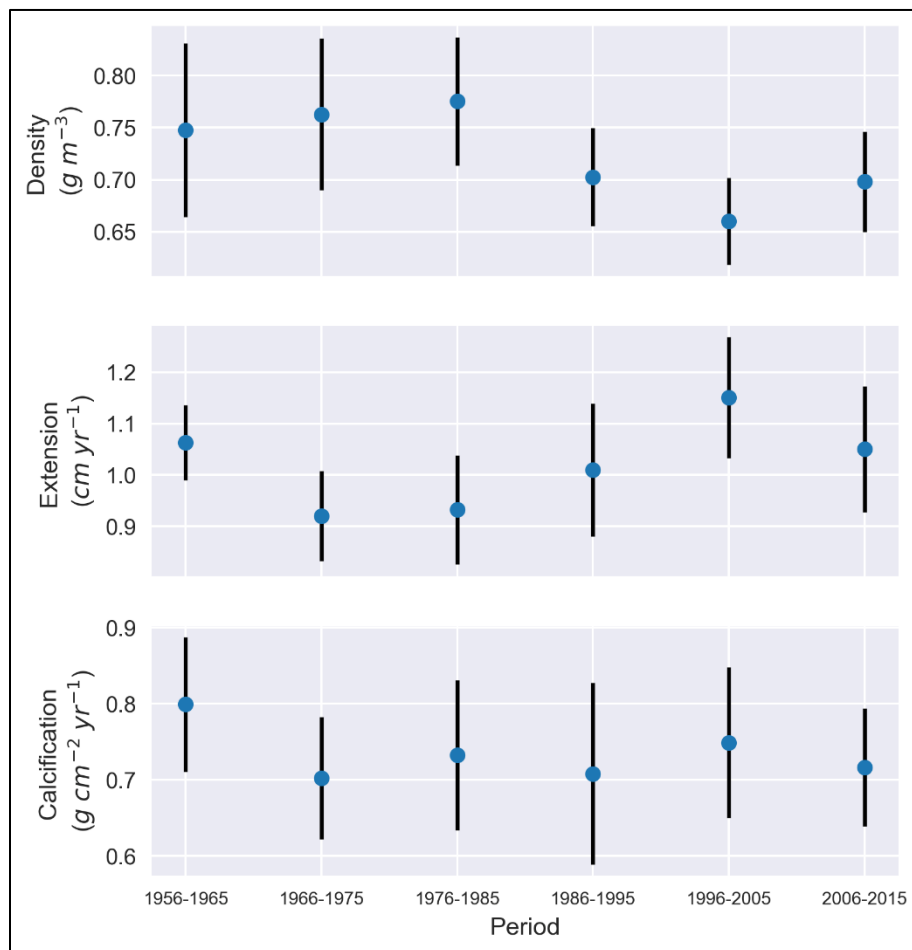
PROMEDIOS POR PERIODOS DE 10 AÑOS (DATOS NO NORMALIZADOS):


FIG. 8. PROMEDIO ($\pm$ STD) DE CRECIMIENTO CORALINO POR DECADA ($n=10$). ELABORADO A PARTIR DEL PROMEDIO GENERAL ($n=4$).

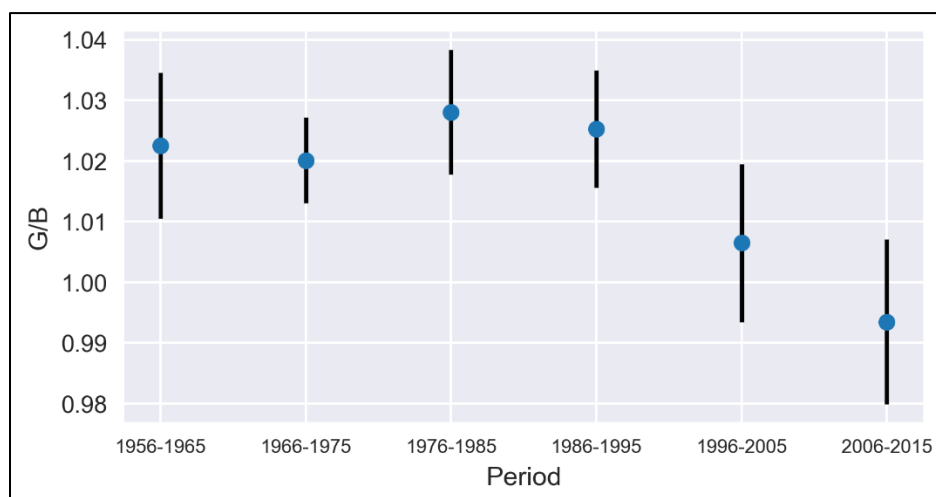


FIG. 9. PROMEDIO ($\pm$ STD) DE LUMINISCENCIA POR DECADA ($n=10$). ELABORADO A PARTIR DEL PROMEDIO GENERAL ($n=4$).

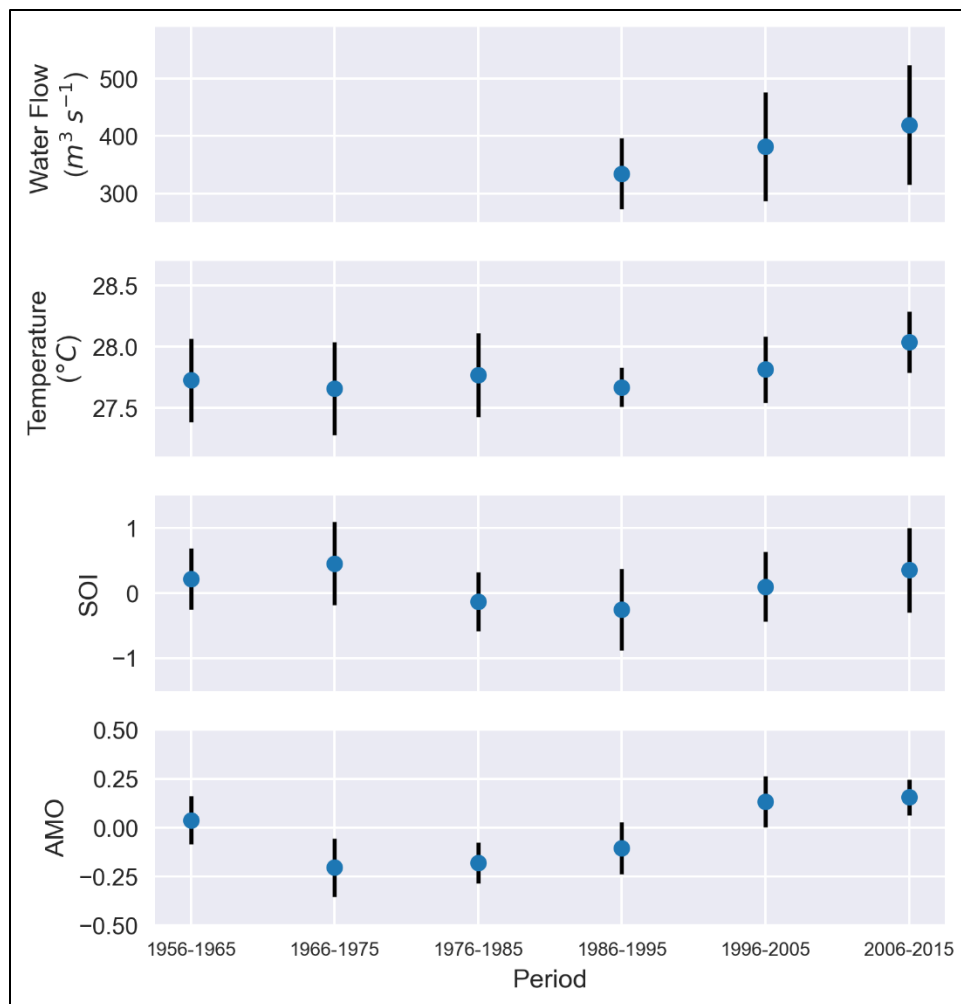


FIG. 10. PROMEDIO ($\pm$ STD) DE VARIABLES AMBIENTALES POR DECADA (n=10).